Week 09: Midterm preparation

Learning outcomes: CLO1, CLO2, CLO3

Lesson

- The midterm covers Weeks 1–8. Your instructor gives the time, rules and allowed materials.
- These tasks are practice, not the exam. Try them without notes first.
- For each query, know: what one row means, the join keys, the filter and the sort.

Worked example

Run `example.sql` (how (`setup/README.md`)), then answer in `work.sql`.

Tables for this week

Read `cambodia`, write `student_work`. All columns are `NOT NULL`. Samples are the first two rows by code.

`cambodia.province`

Column	Type	Key	Meaning
`province_pcode`	`CHAR(4)`	PK	Province code
`province_name`	`VARCHAR(100)`		Province name (English)
`area_km2`	`DECIMAL(12,1)`		Area in km², one decimal place

Sample rows:

province_pcode	province_name	area_km2
KH01	Banteay Meanchey	6145.9
KH02	Battambang	11869.0

`cambodia.district`

Column	Type	Key	Meaning
`district_pcode`	`CHAR(6)`	PK	District code
`district_name`	`VARCHAR(100)`		District name (English)
`area_km2`	`DECIMAL(12,1)`		Area in km², one decimal place
`province_pcode`	`CHAR(4)`	FK	Province code

Sample rows:

district_pcode	district_name	area_km2	province_pcode
KH0102	Mongkol Borei	807.2	KH01
KH0103	Phnum Srok	761.7	KH01

cambodia.commune

Column	Type	Key	Meaning
commune_pcode	CHAR(8)	PK	Commune code
commune_name	VARCHAR(100)		Commune name (English)
area_km2	DECIMAL(12,1)		Area in km², one decimal place
district_pcode	CHAR(6)	FK	District code

Sample rows:

commune_pcode	commune_name	area_km2	district_pcode
KH010201	Banteay Neang	52.0	KH0102
KH010202	Bat Trang	40.3	KH0102

Join keys: `district.province_pcode = province.province_pcode`; `commune.district_pcode = district.district_pcode`.

example.sql lists the `cambodia` columns from `information_schema.COLUMNS`.

More: data guide (`data/README.md`).

Lab tasks

1. List province names and areas by area, then code.
2. Count districts per province, keeping provinces with none.
3. Find districts larger than their province's average district.
4. Draw the three-table ERD; explain PKs and FKs.
5. Explain how to normalize the Week 8 report. Create `student_work.w09_province` from `cambodia.province`, delete one row in a transaction, then roll back.

Homework

1. After feedback, write your mistakes and corrected queries in `reflection.md`.
2. Make a one-page study sheet.

Capstone sprint

No sprint. Keep Checkpoint 2 committed and list your next tasks by priority.

Submit

Answers in `work.sql`, reflection in `reflection.md`, output in `results/`. Commit and push. See SUBMISSIONS.md.